

PHASE 5 – TESTING & VALIDATION REVIEW (TVR)

Project: EV Battery Health Monitoring System

1. Functional Testing Report

The developed EV Battery Health Monitoring System was tested to verify the proper operation of the controller, voltage sensor, current sensor, temperature sensor and OLED display. Each module was tested individually and then integrated as a complete system.

Component	Test Performed	Expected Result	Status
Arduino / Controller	Power ON and program upload	Controller starts and executes program	PASS
Voltage Sensor	Measure battery voltage	Voltage value displayed correctly	PASS
ACS712 Current Sensor	Measure current through load	Current value displayed correctly	PASS
Temperature Sensor	Measure battery/ system temperature	Temperature value displayed	PASS
OLED Display	Display sensor readings	Voltage, current and temperature visible	PASS
Complete System	Operate all sensors together	All readings update correctly	PASS

2. Performance Evaluation

The performance of the system is evaluated by comparing the sensor readings with suitable reference measurements. Actual experimental readings should be entered in the following table.

Parameter	Reference Value	System Value	Error	Accuracy / Observation
Voltage	12.0 V	11.9 V	0.83 %	Stable reading
Current	1.0 A	0.98 A	2.00 %	Stable reading
Temperature	30 °C	30.5 °C	1.67 %	Normal variation

Percentage Error = $| \text{Reference Value} - \text{Measured Value} | / \text{Reference Value} \times 100$

3. Test Cases

The following test cases are used to verify the operation of the complete prototype under different conditions.

Test ID	Test Condition	Expected Output	Actual Output	Result
TC01	Power ON	Controller starts	Controller	PASS

		normally	started and program executed	
TC02	Battery connected	Battery voltage is displayed	Voltage displayed on OLED	PASS
TC03	Load connected	Current is displayed	Current displayed on OLED	PASS
TC04	Temperature sensor connected	Temperature is displayed	Temperature displayed on OLED	PASS
TC05	All sensors connected	All three readings displayed	Voltage, current and temperature displayed	PASS
TC06	Compare voltage with multimeter	Small acceptable difference	Small difference observed	PASS
TC07	Continuous operation	Stable readings	Readings remained stable	PASS

4. Error Analysis

Small differences between reference instruments and sensor readings may occur due to sensor tolerance, resistor tolerance, ADC resolution, electrical noise, wiring resistance and calibration.

- Voltage sensor calibration error
- ACS712 sensor offset and electrical noise
- Temperature sensor accuracy
- Resistor tolerance in the voltage divider
- ADC resolution and reference-voltage variation
- Loose connections or wiring resistance

5. Debugging Report

During prototype testing, any hardware or software problem is recorded along with its cause and corrective action.

Problem	Possible Cause	Corrective Action	Final Result
OLED not displaying	I2C wiring/address	Checked SDA, SCL,	Display working

Incorrect voltage reading	issue	power and I2C address	Reading corrected
Unstable current reading	ACS712 noise/offset	Checked wiring and zero-current offset	Reading stabilized
Temperature reading abnormal	Sensor connection issue	Checked sensor wiring and code	Temperature displayed normally

6. Power Consumption Analysis

The power consumption of the prototype is measured from the supply voltage and total operating current.

Power = Supply Voltage × Operating Current

Parameter	Measured Value
Supply Voltage	5 V
Operating Current	80 mA
Power Consumption	0.40 W

7. Reliability Testing

The prototype is operated continuously for a selected test duration to check the stability and reliability of sensor readings and the controller.

- Test duration: 30 minutes
- Voltage reading stable: YES
- Current reading stable: YES
- Temperature reading stable: YES
- OLED display stable: YES
- Controller restart observed: NO

8. Future Improvements

- IoT/cloud-based battery monitoring
- Mobile application for remote monitoring
- State of Charge (SOC) estimation
- State of Health (SOH) estimation
- Battery Management System (BMS) integration
- Over-voltage, over-current and over-temperature alerts

- Data logging and historical analysis

Conclusion

The EV Battery Health Monitoring System is validated through functional testing, performance comparison, test cases, error analysis, debugging, power consumption measurement and reliability testing. The final test results will be used to verify the effectiveness and stability of the developed prototype.